

5. FMEA and Fault Tree Analysis

This section examines how faults in the knee-joint architecture can contribute to HE-01: hazardous unintended knee motion while the joint is load-bearing. The analysis is deliberately focused on the failure paths that matter to the demonstrator rather than attempting a complete production FMEA or FMEDA.

5.1 Analysis Scope

The analysis covers failures across sensing, communication, control, power electronics, mechanical transmission and the safety-reaction path. The aim is to understand how a local failure can propagate to hazardous joint behaviour and where detection or mitigation can interrupt that propagation.

A failure mode is retained in the portfolio analysis only when it contributes directly to HE-01, affects the ability to detect HE-01, or affects the ability to mitigate it. This keeps the analysis aligned with the project scope.

5.2 FMEA Method

The focused FMEA uses the following reasoning chain:

Failure Mode → Local Effect → System Effect → Detection → Fault Reaction

The distinction between local and system effect is important. The local effect describes what changes around the failed item. The system effect describes how that change can influence the knee joint or contribute to the hazardous event.

5.3 FMEA Observations

- Sensor faults can be hazardous even when the sensor output remains plausible.
- Incorrect current feedback can result in either excessive torque or unintended torque reduction, depending on the failure direction.
- Motor-side and joint-side encoders provide different information and therefore support useful cross-checking across the gearbox.
- A local controller failure is more critical when the same controller is also required to execute the only available safety reaction.
- Power-stage faults can cause physical torque to differ from software intent, so monitoring only commands is insufficient.
- Mechanical transmission faults remain safety-relevant even when all electronics operate correctly.
- The safety monitor and independent torque-disable path must themselves be considered in the failure analysis.

5.4 Fault Tree Analysis Approach

The FTA complements the FMEA by starting from the hazardous system behaviour and reasoning backward toward possible causes. The tree is intentionally small and is used to show the main causal structure rather than quantify top-event probability.

5.5 Top Event

TE-01 — Hazardous unintended knee motion while the joint is load-bearing.

5.6 Fault Tree Structure

The top event is considered to require both a hazardous actuator behaviour and failure of the available safety mitigation. The initiating behaviour is then divided into three broad causal paths.

Node	Meaning / Example Basic Events
A	Incorrect command is generated or accepted. Examples: corrupted/stale communication; local controller generates an incorrect torque/current request.
B	Actuation differs from the intended command. Examples: incorrect PWM/current regulation; gate-driver/inverter fault; gearbox/coupling transmission fault.
C	Feedback or control-state information is incorrect. Examples: joint encoder fault; motor encoder fault; incorrect current feedback.
D1	Safety monitor fails to detect the unsafe condition.
D2	Independent torque-disable path fails to stop or inhibit hazardous torque.

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5.7 FTA Logic and Interpretation

The tree is structured so that A, B and C are alternative initiating paths, while D represents failure of mitigation. This avoids treating every component fault as a direct top-event cause.

For example, an incorrect current command does not automatically create the top event if the safety monitor detects the deviation and the independent reaction path successfully limits torque. Conversely, failure of the safety monitor alone does not create hazardous knee motion unless an initiating fault or abnormal actuator behaviour also occurs.

This distinction is important because it shows that the safety architecture is expected both to reduce the probability of hazardous behaviour and to interrupt its propagation when it occurs.

5.8 Link to Safety Requirements

The FMEA and FTA support the requirements derived in Section 4. The main relationships are:

- FSR-01 / FSR-02 address hazardous joint-velocity deviation and the required reaction.
- FSR-04 / FSR-05 address unintended motor-current behaviour and torque-generation faults.
- FSR-06 addresses inconsistency between motor-side and joint-side motion.
- FSR-07 / FSR-08 address invalid or unavailable robot-level commands.
- FSR-09 addresses the possibility that the normal control path itself is faulty.
- FSR-10 addresses invalid or unavailable safety-relevant sensing.

5.9 Limitations

This is a qualitative system-level analysis. It does not include component failure rates, FIT data, diagnostic coverage calculations, quantitative probability evaluation, or a complete dependent-failure analysis. These would require a more detailed hardware design and reliability data.

The purpose of this analysis is to demonstrate the link between architecture, failure propagation, safety mechanisms and requirements for the selected hazardous event.

5.10 Related Project Artifacts

- 05_FMEA.xlsx — full focused FMEA working table
- 05_fault_tree.drawio — editable fault-tree source
- 05_fault_tree.png or .svg — exported fault-tree figure for GitHub and the final report